

Justin Weonjun Sohn

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks: React, React Native, Next.js

Tools: Node.js, Express.js, Bash, Git, Docker, RESTful APIs, Jira, Pytest

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase, Prisma ORM

WORK EXPERIENCE

TruAbutment

Jan 2025 - Apr 2025

Quality Assurance Engineer

- Drafted and maintained over 20 test documents per release cycle, standardizing defect traceability and improving audit readiness, which reduced lookup time by 30%.
- Conducted root cause analysis on recurring defects, identified systemic issues, and implemented process improvements, reducing defect recurrence by approximately 20% across release cycles.
- Created and maintained test cases, configuration records, and defect logs to ensure product stability before deployment.
- Designed structured test cases with measurable success criteria and applied controlled A/B testing to isolate bugs, detect defects early in the QA cycle, and decrease deployment bugs by 15%.

California State University, Fullerton

Aug 2023 - May 2024

ACM Tutor

- Provided one-on-one tutoring in HTML/CSS and core computer science concepts to over 20 students, creating tailored problem sets and live debugging sessions, which increased average participant grades by 7%.
- Coached students through debugging and code optimization exercises, resulting in cleaner codebases with fewer than half the runtime errors compared to initial drafts.
- Explained technical concepts clearly to non-technical audiences.

PROJECTS

MP4 to Sheet Music Transcription Automator | *Python, basic-pitch (Spotify), librosa, pretty_midi, ffmpeg, MuseScore CLI*

- Developed a modular Python CLI pipeline that converts piano MP4 recordings into PDF sheet music through four independently maintained processing stages: audio extraction, neural network transcription, MIDI processing, and sheet music rendering. Integrated Spotify's basic-pitch machine learning model for note onset and pitch detection from WAV audio, adjusting onset and frame confidence thresholds above default values to reduce phantom note artifacts in the output.
- Implemented beat-aligned sixteenth note quantization using librosa's beat tracker, aligning detected notes to a dynamically generated rhythmic grid and filtering short artifact notes to improve sheet music readability for downstream rendering.

VCT Discord Bot | *Python, asyncio, httpx, pytest, Discord.py*

- Built a Discord bot that scrapes live Valorant Champions Tour match results and standings by integrating with vlr.gg's API, using asyncio and httpx to handle concurrent requests and response caching to reduce redundant fetches.
- Implemented a modular architecture separating data fetching, command routing, and automated posting logic, making it straightforward to extend with new commands or swap data sources.
- Created test coverage for core modules using pytest with mocked API responses to validate error handling and cache behavior.

Mumble – Real-time Communication App | *JavaScript, Node.js, Express.js, EJS, Socket.io, MongoDB*

- Designed a three-tier role architecture (User, Channel Admin, DB Admin) to support access control, enabling secure channel ownership and administrative moderation.
- Diagnosed Socket.io bottlenecks in user-to-user connections and implemented message queuing, reducing average message delivery latency by 15% for concurrent users.
- Delivered a production-ready product to stakeholders within the three-month schedule, meeting team deliverables and addressing stakeholder concerns.

EDUCATION

California State University, Fullerton

May 2024

Bachelor of Science, Computer Science | Dean's Honor List

Association of Computing Machinery (ACM) – Member